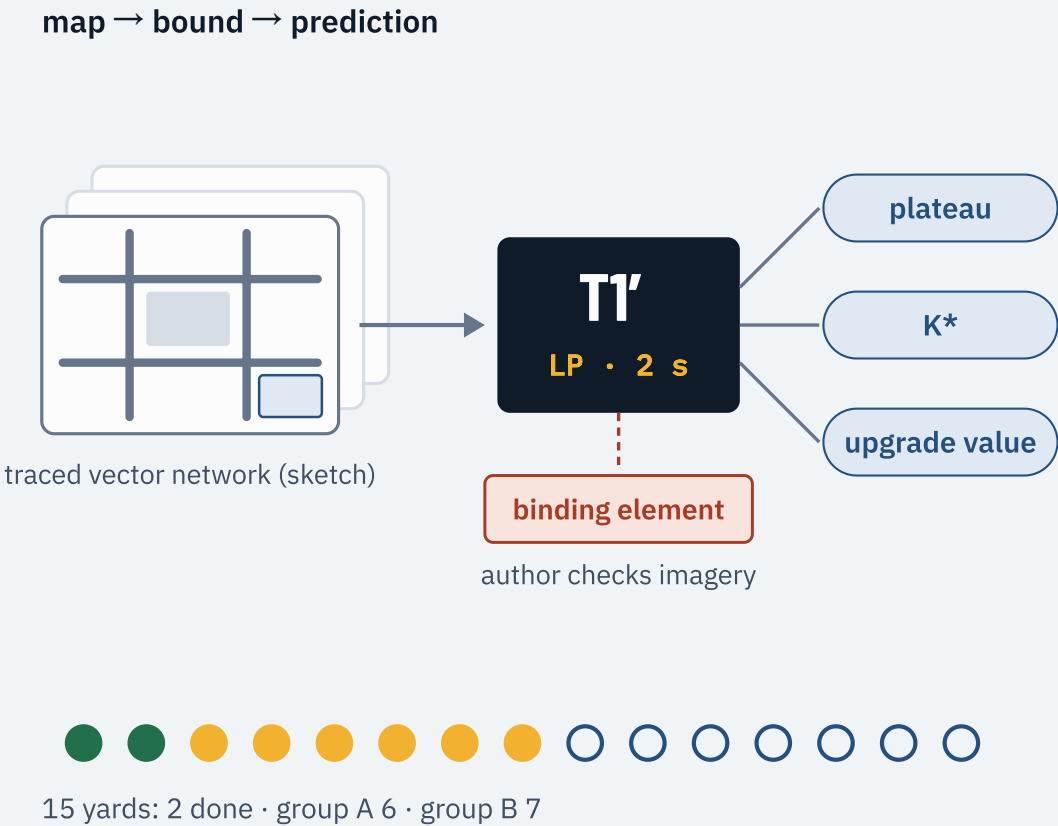


One map, a two-second LP: can it predict a new shipyard's capacity?

Testing across shipyards: the network bound computed from the map, with its binding elements, as a cross-yard predictor (C5, H6)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts (working title)

Chunrui Na
Harbin Engineering University · Nanyang Technological University
2 yards done · group A's 6 being annotated · 15 in all



This sub-study in one minute

1

The question

Do capacity results from two yards carry over? Chapter 6 tests C5: the bound computed from the map, with its binding elements, predicts plateaus, jamming thresholds and upgrade values across yards.

2

The test

The predictor is T1' itself, a two-second LP per yard, with its binding elements; tested by leave-one-out, yards grouped by key element. Topological indicators explain where the bound binds; they do not replace it.

3

Two points so far

Okpo is a dock-road yard, Yantai a door-type yard. Reservation plateaus sit at 36% and 35% of the system T1'; each predicts the other within about 3%. At its own volume Yantai needs 1–2 vehicles; Okpo's peak (16 h) needs 117, inside the network-bound regime.

4

Next

All 20 listed yards have maps and base maps; the author is annotating group A's six. Each gets T1' and its binding-resource reading first, then the author checks the top three elements on imagery, then simulations. Group B follows at eight yards; 15 in all.

Bound first, binding element next, simulation last: screening a new yard starts with a two-second LP.

Sources: research manifesto v1 (thesis-core tag manifesto-v1), C5 and H6; review and follow-up guidance (9 Oct 2026), §3 question 6 and §4.3; numbers in box 3 on pages 6–10 (thesis-core a882742: T22, T23); box 4 from the yard list (10 Oct 2026), §2–3.

The question: which yards do the earlier results carry over to?

C5	Transferable The network bound computed directly from the map, and its binding elements, predict plateaus, jamming thresholds and upgrade values across shipyards
H6 Cross-yard prediction	
Hypothesis	The network bound and binding elements computed directly from the map predict the simulated plateau, K^* and upgrade values across shipyards. Topological indicators serve to explain why the bound binds where it does.
Test	Leave-one-out test; classification by key element: dock-road single-lane type, door type, ...
If it holds	A two-second linear program is enough to screen a new shipyard.
If it fails	What transfers is the bound tool and the key-element classification themselves.

Where chapter 6 sits in the thesis

Ch. 3 Capacity	$T1'$, the capacity interval [best deadlock-free rule, $T1'$] and the K^* criterion: the predictor and the first two targets
Ch. 4 Orchestration	Pushes the interval's lower edge up: plateaus move with the rule, so each rule is read separately
Ch. 5 Co-design	Dual prices and upgrade simulations: the third target, upgrade value
Ch. 6 Many yards	Which yards these results carry over to: prediction errors, the key-element classification and a screening procedure

H6 pays either way: if it holds we gain a screening tool for new yards; if not, the bound tool and the key-element classification are themselves what transfers.

The predictor is T1' itself, not a topology regression on 15 samples

Regressing topological indicators on 15 yards

- Fifteen samples support one or two predictors at most; easy to overfit
- Traffic rules and task flows sit between indicators and capacity

T1' as the predictor

- From the map in two seconds per yard: an occupancy LP with width, crane and window rows
- System T1': the task stream's expected occupancy, no sampling noise (Okpo 1,736, Yantai 1,298 a day; T23)
- Upper edge of the interval (C1) for rules meeting (A1)–(A3), loaded routes fixed; the route-free T1'' holds for any routing
- Dual prices name the binding element: where to look first on imagery
- Structurally among the closest: railway capacity without a timetable (Jensen et al. 2020, branch and bound)

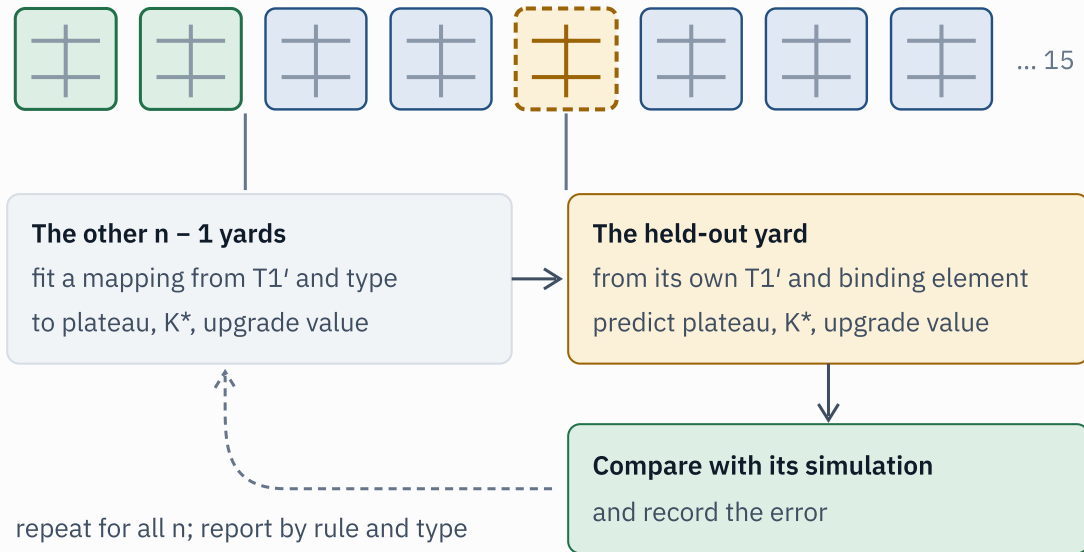
Topology of the two yards: to explain, not to replace

	OKPO	YANTAI
Roads / centreline	721 / 40.3 km	140 / 8.6 km
Junctions / entrances / stops	334 / 218 / 421	58 / 32 / 140
Independent cycles μ / meshedness α	82 / 0.064	14 / 0.056
Dock-bound minimum cut (sum of widths)	62 m	137 m
Dock entry degree (max)	5	5
→ system T1' (a day)	1,736	1,298
→ binding element	dock-road single-lane	door type
→ K*(20%)	50	25
→ reservation plateau (a day)	627	455

Similar loop density, yet K* differs twofold and the binding element differs in type. The indicators explain why it binds there: a narrow dock-bound cut at Okpo, a door at Yantai.

Sources: review §3 question 6; indicators per the T19 report §4.3 (meshedness after Buhl et al. 2006); system T1' and T1'' in the T23 report §0–2 (a882742); other numbers from the t18–t20 summaries and bounds.csv; Jensen et al. 2020 checked in full text (T24). Conditions: whole-route reservation, saturated, MY-B mix, whole-segment resources, Okpo rev 1825, Yantai r396. T1' and T1'' proofs: first drafts the author is rewriting (manifesto §4).

The test: leave-one-out, with yards grouped by key element



Two OD samplings (by entity, by stop) are each yard's standard sensitivity (T19). New types come from the new yards' dual prices, not fixed in advance; each type records the resource granularity it was read under.

Three targets

Plateau

The interval's lower edge, read per traffic rule (chapters 3, 4)

K^*

Network takes over: marginal gain first below 20% of free flow (H1, chapter 3)

Upgrade value

Dual-price prediction against upgrade simulations (chapter 5; T25, after T27)

Two key-element types so far (a third reading on page 8)

Dock-road single-lane · Okpo

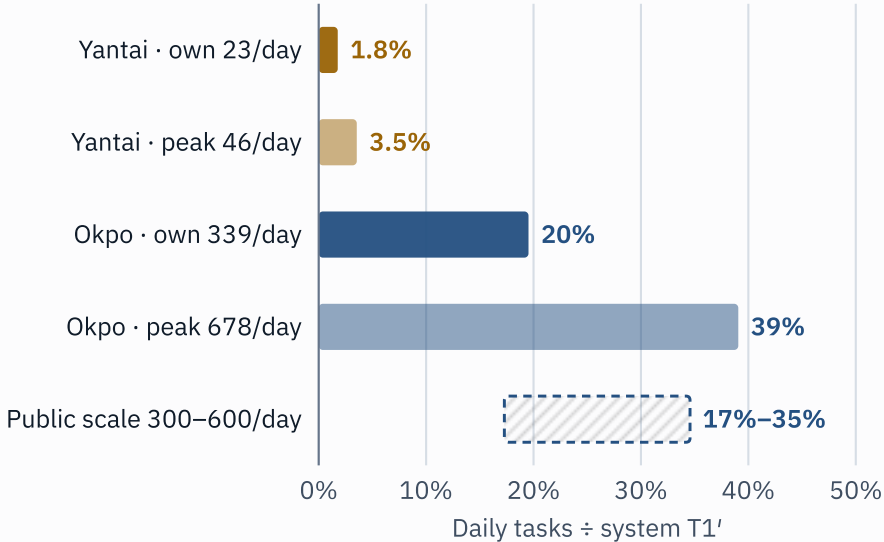
Road 161, dock 1's only stopping road (9 m × 1,095 m); T1' binds there in 5 of 10 seeds, on the narrow road 010-6 in 4.

Door type · Yantai

Building 014's two doors (paint shop, presumed): each takes about 7% of all tasks and is held about 10 min per load; T1' binds there in 9 of 10 seeds.

The first two data points: Okpo and Yantai fall into two key-element types

	OKPO	YANTAI
Map version (of these results)	derived map rev 1825	r396
System T1' (a day; 10-seed mean in brackets)	1,736 (1,720)	1,298 (1,252)
Capacity interval [best deadlock-free rule, T1']	[627, 1,720]	[455, 1,252]
Reservation plateau ÷ system T1'	36%	35%
K*(20%) (range)	50 (50–60)	25 (25–25)
Where T1' binds (of 10 seeds)	road 161 ×5 · road 010-6 ×4 · road 026 ×1	building 014's two doors ×9 · a door of building 007 ×1
Type	dock-road single-lane	door type
Dock threshold (safe release, whole road, K = 20 → 150)	0.98 → 0.59	0.93 → 0.38
Own daily tasks ÷ system T1' (mean–peak)	339–678: 20%–39%	23–46: 1.8%–3.5%
Vehicles needed at own mean (16 h, reservation) vs K*	27 < 50	1 < 25

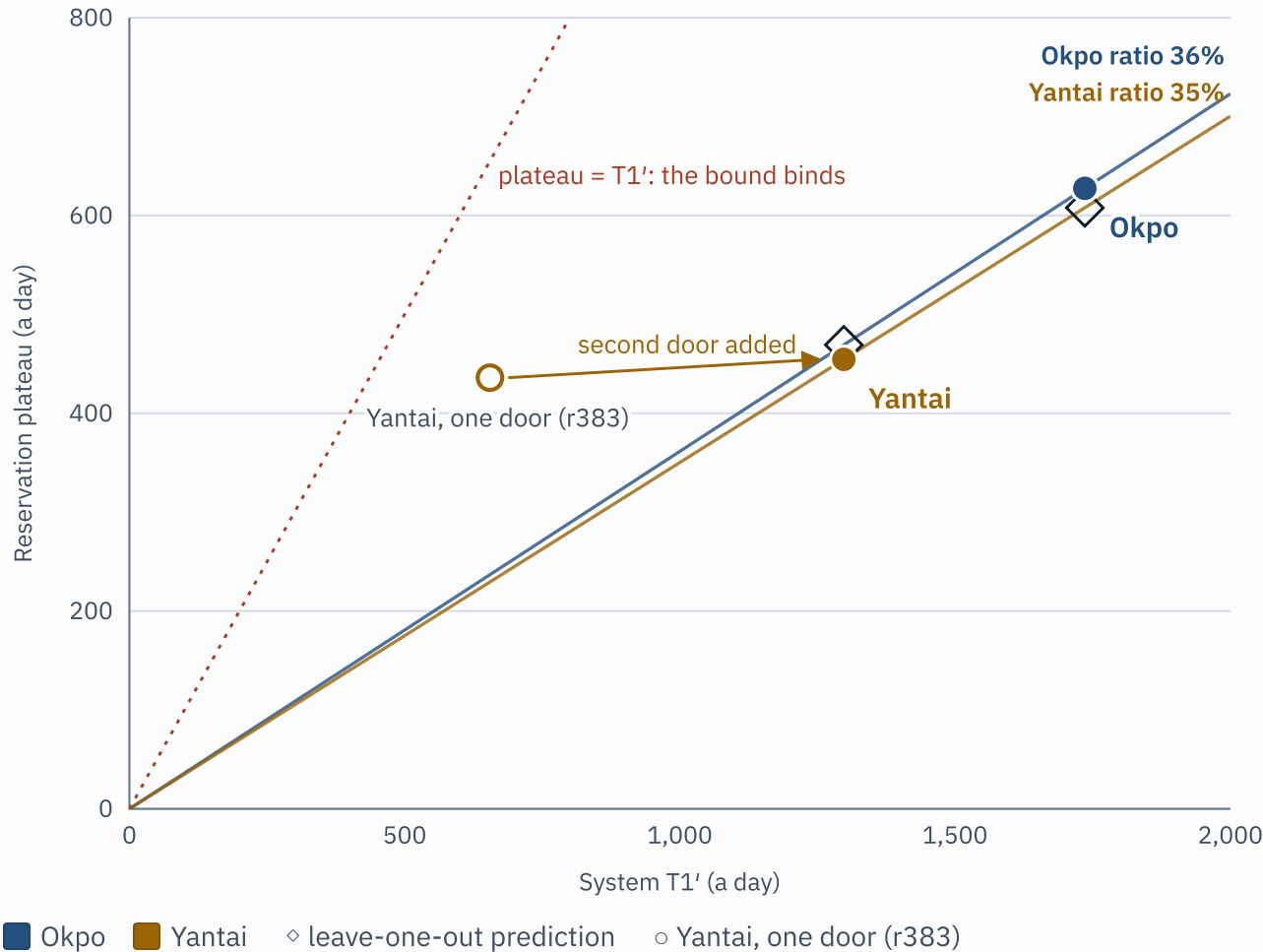


■ Yantai ■ Okpo
▨ public scale of large Korean yards, not Okpo itself (Yim 2008; Shen et al. 2018)

Density criterion: daily tasks ÷ system T1' (H1). One folding for both yards: steel ÷ 100 t × 10 moves ÷ 300 days. Okpo 1,018,067 t a year → 339 a day (294–363 across years; Hanwha's direct anchor 350); Yantai 68,512 t → 23 a day.

Sources: thesis-core a882742 (T18–T23, each reviewed with an independent review): bounds.csv of Okpo results/a0ab16fceb94 and Yantai results/b82aa19b22af, the t18, t19, t20 summaries, t20_demand.csv; T22 report §3.1, 4, 6, 7 (t22_summary.json, t22_demand.csv); T23 system.csv. Okpo's daily volume is T24's same-method folding (library notes/occupancy_anchors_20261009.md §2.5, 6); 100 t blocks after Zhang 2025, 10 moves after Wang 2017; public scale 300–600 a day in Yim 2008 and Shen et al. 2018. Conditions: whole-route reservation (the best deadlock-free rule among those tested), saturated, MY-B flow mix, plateau = mean over K = 100–150, whole-segment resources, 10 seeds; threshold conditions on page 9.

First leave-one-out reading: plateaus at 36% and 35% of the system T1'



36% / 35%

Reservation plateau ÷ system T1': Okpo / Yantai

±3.2%

Error when one yard's ratio predicts the other's plateau (±0.5% with the 10-seed sampled T1')

67% → 36%

Yantai before and after building 014's second door (10-seed mean):
T1' 655 → 1,252, plateau 436 → 455

Two points only show how the test runs. The predictor is the system T1': the sampled T1' runs low in expectation (T23's draft Jensen proposition), and without that noise the error is about 3%. The ratio moves with the traffic rule, so chapter 6 reads it per rule and per type. Near 1 the bound binds and road changes pay; far below 1, orchestration first: Okpo's erection mix, 0.87 and 0.53 with blocks, marks the C3–C4 boundary.

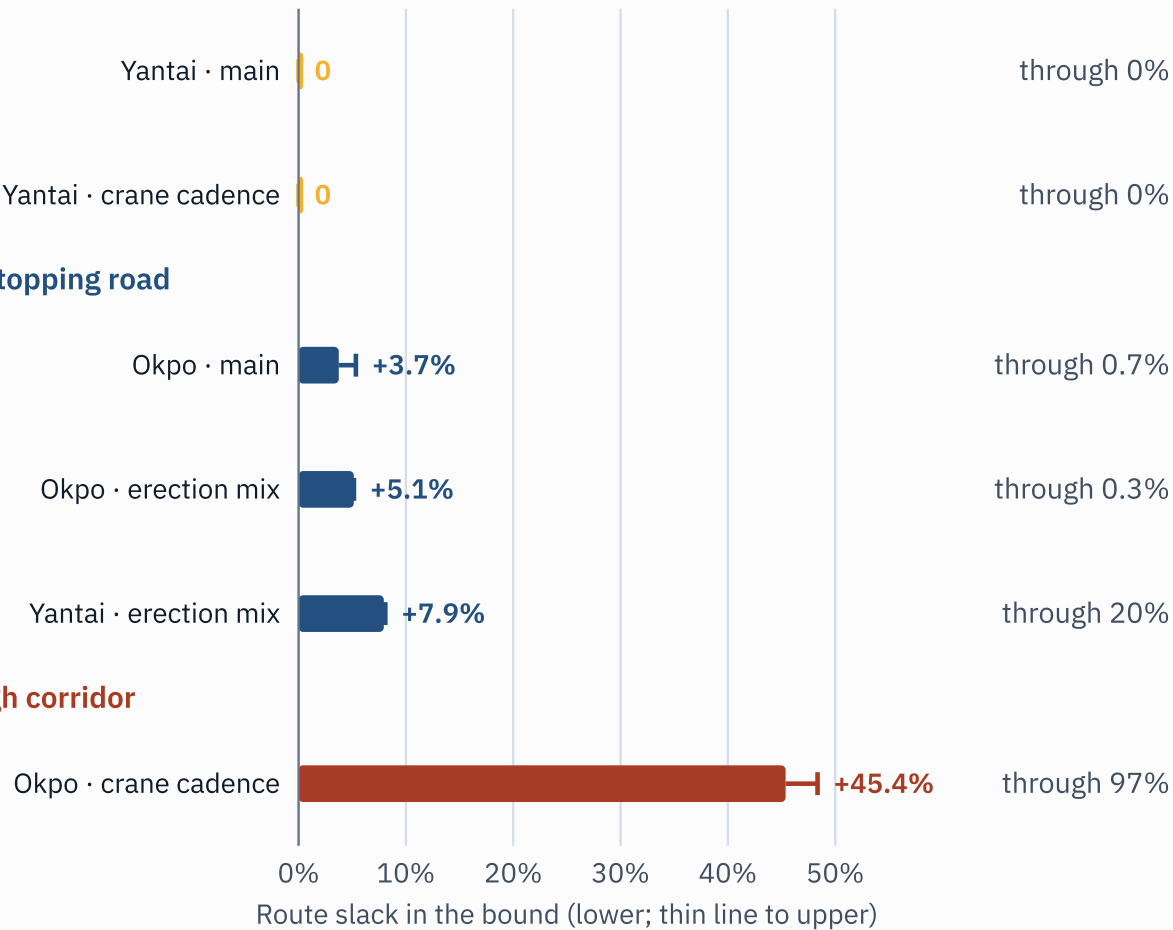
Sources: thesis-core a882742: system T1' from T23 system.csv (results/97d667d16273, e7f94238b96d), 10-seed means from bounds.csv, reservation plateaus from t18_summary.json and t20_summary.json (mean over K = 100–150); Yantai's one-door setting from results/t20_report_c12289dfda9d/t20_r383_r395.csv (10-seed mean); the erection-mix ratios in the T22 report §4. Leave-one-out: one yard's plateau ÷ system T1' times the other yard's system T1'. Conditions as on page 6; the C3–C4 boundary is part of C4 in research manifesto v1, §2.

The binding resource, read three ways: measurable type signatures

Entrance

Dock stopping road

Through corridor



Entrance · door type

Yantai building 014: no through traffic, zero slack by construction; every route is the same. Only the door moves the bound (T25's door granularity).

Dock stopping road · dock-road type

Okpo road 161, Yantai road 007: 0.3–20% through traffic, slack +3.7% to +7.9%, from stops entered at the other end. Granularity, such as blocks, is the first-order lever.

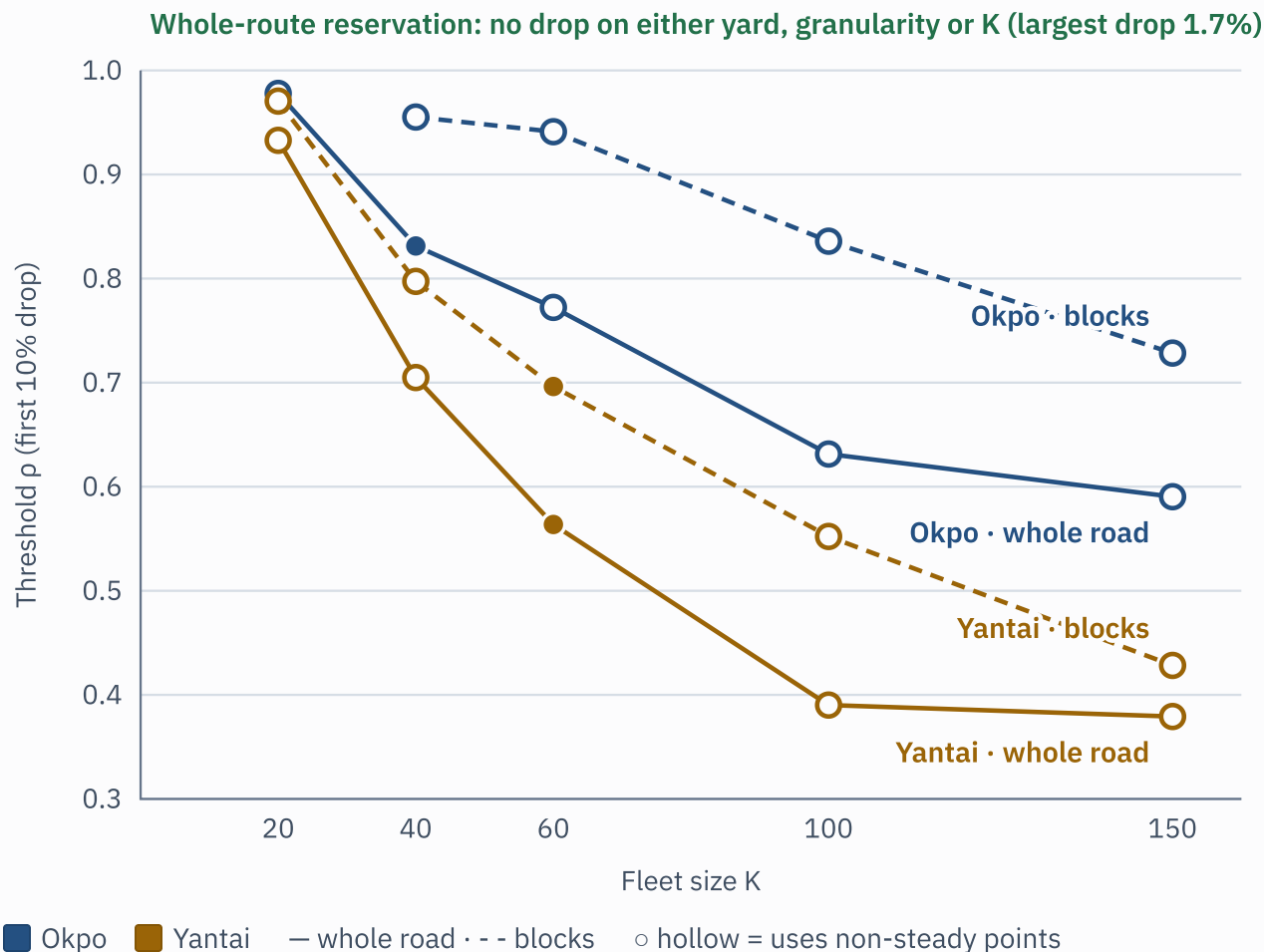
Through corridor · newly read

Okpo road 010-6 under crane cadence: 97% through traffic; its detours keep 22% spare and add 235 m; slack +45.4% (upper +48.4%). Detour orchestration can recover it (chapter 4).

Both signatures (through-traffic share, spare occupancy on the detour) come from the bound's solution, without simulation: H6's types are written in them. Six scenarios so far, 1–3 points per type; T25's upgrade scenarios add more.

Sources: thesis-core a882742, T23 report §0, 2.2, 2.6 (suggested wording); readout.csv and route_free.csv in results/97d667d16273 and e7f94238b96d. Conditions: system T1' (loaded moves on shortest paths) and T1'' (any route), steady state, saturated, Okpo rev 1825, Yantai r396. Route slack is a difference of bounds, not an achievable gain: lower = back-substituted feasible value – T1', upper = T1'' – T1'; the independent review found no counterexample to T1'' (reviews/T23.md).

Second target: the dock threshold, same mechanism, yard-specific values



1

Wait off the road: no drop

Whole-route reservation holds on both yards, both granularities, $K = 20-150$ and p up to 0.99 (largest drop 1.7%).

2

Wait on the road: the threshold moves left with fleet size

Safe release (no teleport), whole road, $K = 20 \rightarrow 150$: Okpo $0.98 \rightarrow 0.59$, Yantai $0.93 \rightarrow 0.38$; blocks move it right (Okpo, $K = 40$: $0.83 \rightarrow 0.96$).

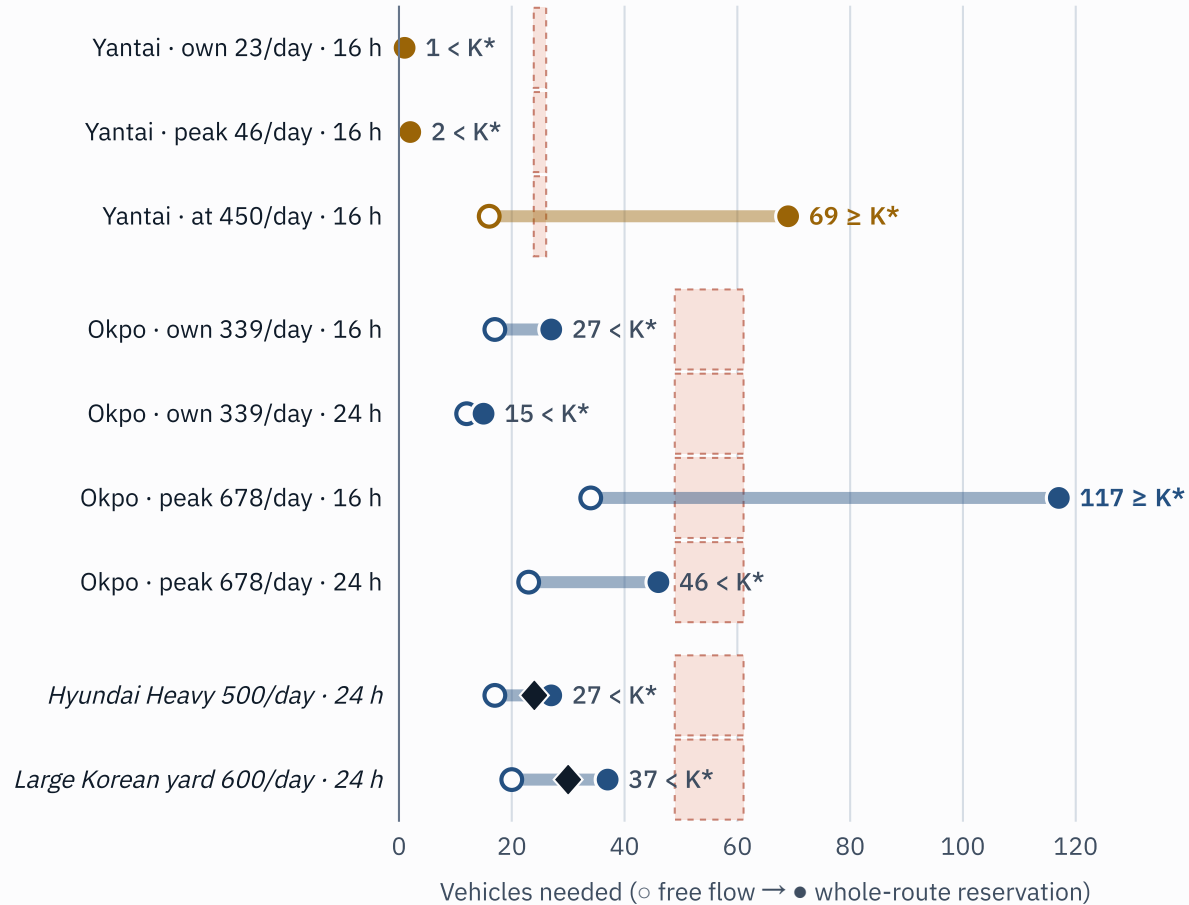
3

The mechanism transfers; the values must be predicted

At every K Okpo's threshold is above Yantai's ($K = 60$: $0.77 / 0.56$). Chapter 6 has to get this number from each yard's key elements; T25's three-resource dock network relates the threshold to the passable room at the dock, a candidate predictor.

Sources: thesis-core a882742, T22 report §3.1, 3.2; results/t22_report_3287d62c7916/t22_summary.json (rho_k). Conditions: crane-cadence p curves, saturated, 8 simulated days (day 1 warm-up), 10 seeds; threshold = the p at which throughput first drops 10% below the lowest p step (Okpo 0.53, Yantai 0.13), interpolated between steps; hollow = uses non-steady points; Okpo blocks at $K = 20$ never drop 10% and are not drawn; Okpo rev 1825, Yantai r396. Safe release = segment requests with a Banker-type safety check, deadlock-free, no teleport.

Either side of H1: vehicles needed at whole-yard volumes against K^*



■ Yantai ■ Okpo ■ K^* range ◆ real fleet

1

Yantai: its own volume is on the other side

Its own 23 / 46 a day need 1–2 vehicles, far below $K^* = 25$; at 450 a day (16 h) it needs 69 and enters the regime too.

2

Okpo: below at its mean, above at its peak

Own 339 a day: reservation needs 27 (16 h) or 15 (24 h), below K^* (50–60). Peak 678: 117 at 16 h, inside the regime; 46 at 24 h, at its lower edge.

3

Real fleets sit between the model's two ends

Hyundai Heavy's 24 vehicles (about 500 a day, 24 h) and a large Korean yard's about 30 (about 600 a day) both sit between free flow and reservation under a 24 h day.

The main caliber (16 h or 24 h, mean or peak) is the author's decision after T27; both are shown here.

Sources: thesis-core a882742, T22 report §6.1, 6.2, 7 (t22_demand.csv); Yantai's own volume from T20 (t20_summary.json, K_{required}). Conditions: main flow mix, whole-segment resources, 4 days, 10 seeds; vehicles needed = smallest integer K whose 10-seed mean throughput $\geq 99\%$ of the daily tasks; K^* range Okpo 50–60, Yantai 25; peak = mean $\times 2$. Field anchors: 박창규 & 서준용 2006 (Hyundai Heavy, 24 h); Shen et al. 2018 (yard unnamed, shifts not stated, read at 24 h); both checked in full text by T24.

Key-element review: the top three elements of T1' , checked on imagery

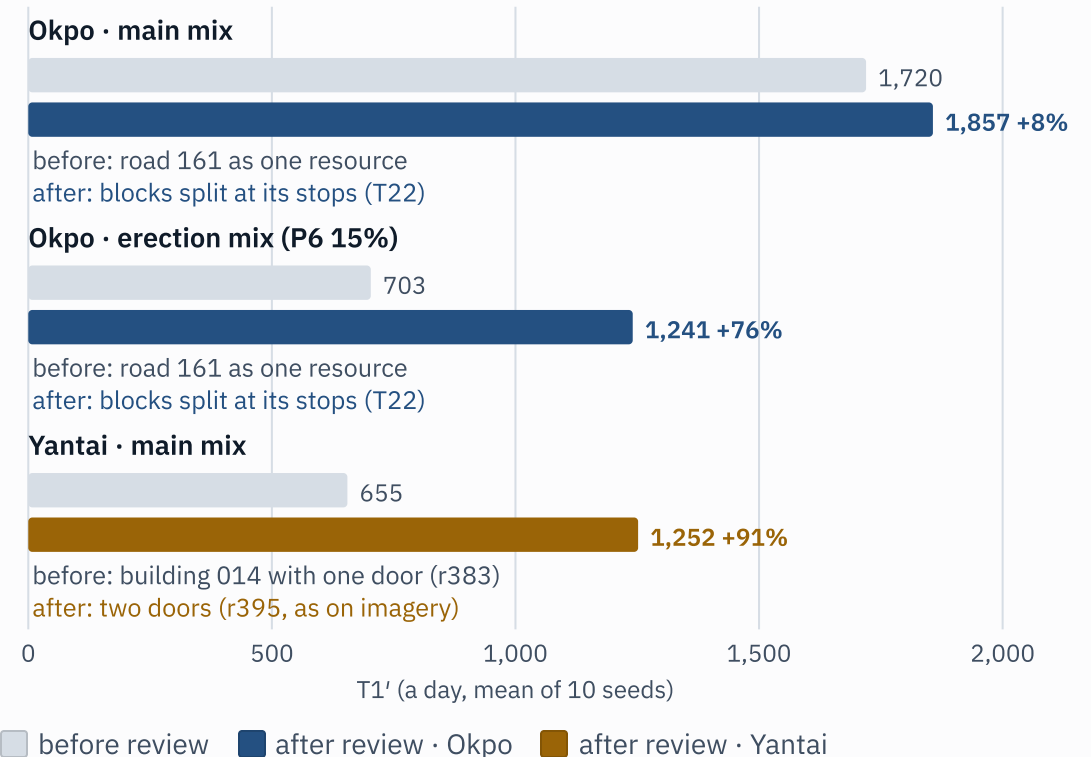
- 1 **List the top three elements by dual price**, with building, entrance name and coordinates, not just a label
- 2 **The author checks three things on the imagery**: road width; the number of doors; how traffic passes roadside stops and doors (while a vehicle works there, can others get in and out, can they pass?)
- 3 **Record it in the map's source field**
- 4 **Choose the resource granularity and recompute T1'**: whole road or blocks, door held or shared

Okpo · road 161

The model held the 1.1 km dock road as one resource. Imagery: neighbouring assembly sites stay reachable while a vehicle works at dock 1. → T22 splits it at the stops: main mix upper edge +8%, lower +0.3%; erection mix upper +76%, lower +8.3%. Blocks are Okpo's main scenario.

Yantai · building 014

In T19 one door was mapped; the review added the second, as on the imagery. On 9 Oct: vehicles can pass at the doors, so holding a door while loading is conservative. → T25 (after T27) adds door granularity.



On both yards the binding element sat on a conservative modelling choice, and the check changed the model. Three elements per yard, no precise annotation: a minimal human check, written into the thesis as an engineering method.

Fifteen yards: 2 done, 6 in group A next, 7 in group B

— Done — Group A: next, reaching eight yards in six countries — Group B: up to 15

Hanwha Ocean Okpo KR LNG carriers, container ships, offshore Flag-experiment yard; dock-road type. Also builds submarines and naval ships, disclosed	CIMC Raffles Yantai CN rigs, FPSOs, wind installation vessels Offshore yard; door type. Public environmental and cleaner-production filings	Chantiers de l'Atlantique FR cruise ships Large European cruise yard; IGN open imagery, the richest OSM	Meyer Werft Papenburg DE cruise ships Covered building docks, a distinct dock type; LGLN open imagery	Imabari Saijo JP bulk and ore carriers, container ships A building dock over 400 m with three large gantry cranes; GSI imagery
Philly Shipyard US Jones Act tankers, container ships Small single-dock yard; NAIP public-domain imagery	New Yangzi Shipbuilding CN container ships, bulk carriers, tankers Large multi-dock yard; reports of a Singapore-listed group	Samsung Heavy Geoeje KR LNG carriers, container ships, offshore On the same island as Okpo; DART filings	Fincantieri Monfalcone IT cruise ships Second European cruise yard, civilian only; Esri 2025 imagery	Oshima Shipbuilding JP bulk carriers Mid-size Japanese bulk-carrier yard; GSI imagery
Namura Imari JP bulk carriers, tankers A third Japanese yard; GSI imagery	CM Jinling Weihai CN ro-pax ferries Small yard with a distinct product	Hengli Heavy Industry CN bulk carriers, tankers, container ships Expanded in 2024; imagery must postdate it	HD Hyundai Samho KR VLCCs, container ships, LNG carriers Large yard; DART filings	K Shipbuilding KR MR tankers, mid-size container ships Mid-size yard; moved up from group C on 10 Oct, replacing HD Hyundai Mipo

Selection looks for spread: dock type, key-element type and the level of daily tasks ÷ T1' matter more than the count.

Source: yard list (10 Oct 2026 version), §2; all 20 listed yards (2 done and 18 in groups A, B, C) have maps and base maps, kept in 20_数据/船厂地图; list and reasons checked against public sources, and the entries named at the end of its §6 are verified together with facility use before inclusion. Country codes: KR Korea, CN China, FR France, DE Germany, JP Japan, US United States, IT Italy.

Selection rules: civilian yards, open data first, wide variety

- 1 **Civilian yards:** naval builders left out (Hudong-Zhonghua, GSI Nansha, HD Hyundai Mipo); Okpo, the finished flag yard, stays, naval business disclosed
- 2 **Open data first:** open-licence orthophotos, else Esri imagery for local tracing; OSM roads as drafts; filings to check facility use and output
- 3 **Wide variety:** size, layout, dock type (open, covered, single, multiple), region, product
- 4 **Checks before inclusion:** imagery from 2022 on and after any expansion; blocks reach the dock by road
- 5 **Choose by key-element type and by daily tasks ÷ T1':** spread matters more than count

Two cautions

- **Philly Shipyard:** Hanwha plans expansion and naval orders; civilian for now on dated pre-expansion imagery, swapped for group C if it turns mainly naval
- **Covered dock halls:** if blocks do not reach the dock by road, rule 4 removes the yard

Group C reserves (5), for any A or B yard that fails a check

- (New Times (Jiangsu) CN) (Shanghai Waigaoqiao CN)
- (JMU Ariake JP) (Tsuneishi Zhoushan (SW Xiushan) CN)
- (Daehan Shipbuilding KR)

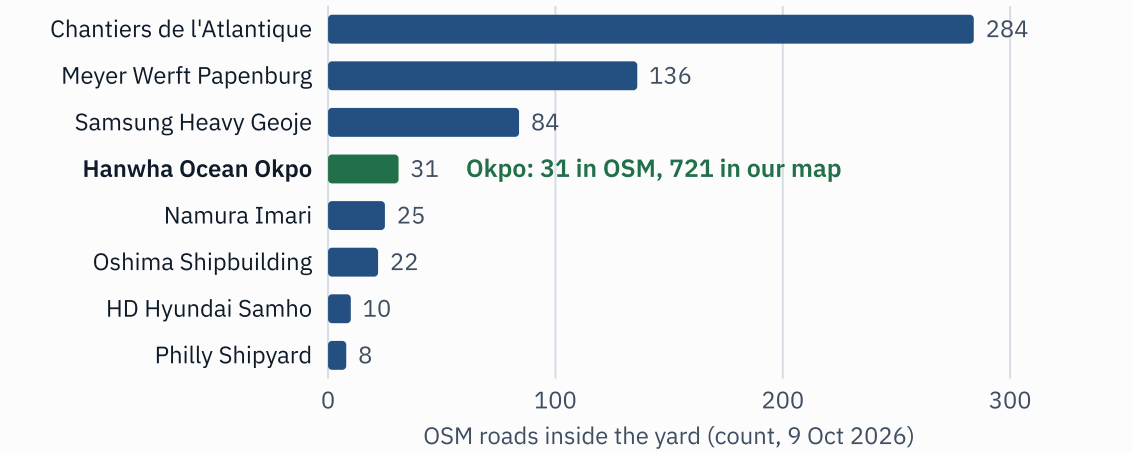
Removed from the list

- **Hudong-Zhonghua, GSI Nansha:** naval shipbuilding
- **Meyer Turku (Finland):** some blocks come by barge from Germany, not by yard road
- **HD Hyundai Mipo (10 Oct):** merged into HD Hyundai Heavy, some docks turning to naval ships; out under rule 1, K Shipbuilding moves up

Open imagery and OSM drafts; figures show only our traced vectors

REGION	IMAGERY	LICENCE AND USE
France	IGN BD ORTHO (about 20 cm)	Licence Ouverte 2.0, credit IGN
Lower Saxony, Germany	LGLN DOP20 (20 cm)	CC BY 4.0, credit “LGLN (year)”
Japan	GSI aerial photo tiles	use with attribution
United States	USDA NAIP (about 60 cm)	public domain
Friuli Venezia Giulia, Italy	IRDAT / Eagle.fvg (2017–2020)	open licence, but older than 2022: not used
China, Korea (existing base maps)	existing TIFs (about 0.25 m)	tracing only, never in figures
Other Chinese and Korean yards, Monfalcone	Esri World Imagery (2022–2025, 0.31–0.5 m)	not open: local tracing only, not redistributed (author, 10 Oct 2026)

Open-licence terms are checked against the originals on inclusion (IGN, NAIP and GSI pending). Capture dates: Philly NAIP 10 May 2022; the Esri yards December 2022 to August 2025 by metadata; the rest pending. Figures show only our traced vectors.

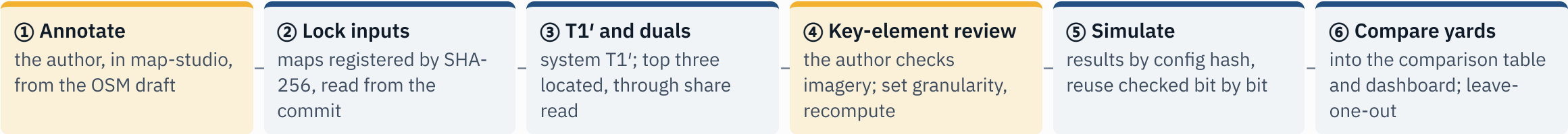


- **All 20 listed yards have maps and base maps** (20_数据/船厂地图): 11 OSM drafts, 3 drafts by extent, 4 blank maps on existing base maps, copies for Okpo and Yantai; in map-studio the author checks, adds roads, measures widths, marks facilities and entrances
- **Data sharing:** OSM-derived parts are shared alike under ODbL
- **Optional accelerator:** MY-A's segmentation extracts roads, yards and buildings from imagery where OSM is sparse

Open the segmentation (MY-A) briefing →

Sources: yard list (10 Oct 2026 version), imagery table of §1, §2 and the base maps of the remaining yards (OSM road counts from Overpass, 9 Oct 2026, © OpenStreetMap contributors, ODbL; Esri dates from World_Imagery_Metadata); tasks.md, multi-yard annotation (10 Oct 2026); review §2.7 item 4 (imagery terms, ODbL). No imagery or network geometry is shown.

Per yard: bound first, then the key-element review, then simulation



STEP	WHO	STATUS	OUTPUT (CHAPTER 6)
Yard list; maps and base maps for 20 yards (20_数据/船厂地图)	curation	done	Table of yards and data sources
Annotating group A's 6 yards	author	running	Multi-yard instance library (vector networks)
System T1', top-3 duals and binding-resource reading per yard; two OD samplings	computation	on each map	Comparison table: T1', binding element, type, through share
Key-element review, then recompute	author	fixed step	Review record (map source field)
Verify facility use and output; daily tasks by T24's same folding	curation, library	per yard	Density chart; vehicles needed against K*
Flag simulations: best deadlock-free rule after T27, T22's calibers	computation	after T1'	Plateau, K*, capacity interval, dock threshold
Topology of 8 yards; order group B by the gaps	curation	after A	Indicator table that explains
Leave-one-out and types: a round at 8, final at 15	—	planned	Main figure: prediction against simulation; type table

Sources: yard list §3; tasks.md, multi-yard annotation; review §1.4 (inputs locked by SHA-256, results by config hash, reuse checked bit by bit) and §4.3–4.4 (chapter 6 plan; weeks 5–8 compute T1' and plateaus for the first yards of group A); yard maps always come from 20_数据/船厂地图 (author, 10 Oct 2026).

What is established, and what is in progress

Established (thesis-core a882742; T24 in the library)

- T1' runs on both yards and no checked point exceeds it; the system T1' removes sampling noise, T1'' holds for any routing (T23)
- Both ends of the capacity interval; the best deadlock-free rule is whole-route reservation (T22)
- Two key-element types, plus three readings of the binding resource
- Plateaus at about 36% / 35% of the system T1'; the dock threshold works the same way on both yards
- Okpo's own 339 a day (T24); the two yards fall either side of H1, real fleets between the model's two ends
- Key-element review is a fixed step; maps for 20 yards, SHA-256 locks, bit-by-bit reuse checks

TASK	CONTENT	WHAT IT BRINGS TO CHAPTER 6
T27 running	Step zero: maps from 20_数据/船厂地图 (Okpo rev 1833); a less conservative deadlock-free rule (Banker plus a step-wise feasibility search, or claims cut at the next safe holding point, theorem T5); re-checks the lower edge, H2's decline and H5	The rule for every yard's simulations; plateau ÷ T1' may rise
T25 after T27	On T27's code and rev 1833: upgrade simulations and the first road–vehicle exchange-rate table; door granularity; block-length curve; three-resource dock network	The upgrade-value target; more points for the three readings; a candidate threshold predictor
T26 queued	Floating-dock scenario; finite holding area S and K _o (S)	Okpo's final layout

After T27, three items go to the author together: H1 part 2's new numbers, part 3's two-layer wording, and the main caliber (16 h or 24 h, mean or peak).

Sources: thesis-core a882742 (T17–T23); T22 report §0 item 8 (per-point T1' checks); T24 and T24b in library commits c881609, e483924 and reviews/T24.md; task contents from the T22–T27 rows of tasks.md (10 Oct 2026; the author set T27 first on 10 Oct 2026). Running and queued tasks are plans, not results.

Risks and mitigations

RISK	MITIGATION
Plateau ÷ T1' moves with the rule and the key-element model (67% for Yantai with one door); T27 may lift the lower edge	Read per rule and type, one column per new rule; two OD samplings as standard sensitivity; review the key element first
The binding element sits on a conservative modelling choice (on both yards)	The key-element review is a fixed step; every reading states its granularity; through share and detour slack say which lever to change
Fifteen samples are few; each of the three readings has 1–3 points	No regression: T1' as the predictor, types by key element, leave-one-out; T25's upgrade scenarios add points on every yard
H1's reading depends on the working day and on mean or peak	16 h and 24 h, mean and peak side by side; the author sets the main caliber after T27; Hyundai Heavy and Shen et al. as field checks
Yards without suitable open imagery rely on Esri imagery; most capture dates still to check	Esri for local tracing only, never redistributed; figures show vectors only; terms and dates checked on inclusion; OSM parts under ODbL
A yard fails after selection (turns mainly naval, or blocks arrive by sea)	Five group C reserves; HD Hyundai Mipo already swapped for K Shipbuilding; Philly and covered halls have rules

Sources: review and follow-up guidance §2.6, §2.7, §5 and §3 question 6; yard list (10 Oct 2026 version) §1–2; T22 report §4, 6, 7, 10, 15 and T23 report §2.6, 7 (thesis-core a882742); T20's r383 → r395 comparison.

Where paper four goes: journals in the school's 2026 list, and a working title

Working title (direction)

Do jamming thresholds transfer across shipyards? Network bounds and binding elements as predictors on public layouts

The direction document's question, reframed around manifesto H6: the bound and its binding element as predictors

FIRST CHOICE · CLASS B

***Ocean Engineering* (class B)**

Engineering methods for shipyard production and transport; the 15-layout library and screening procedure are the contribution

***Journal of Marine Science and Application* (class B)**

Naval and ocean engineering; multi-yard instances and bound-based screening for practice

ALTERNATIVE · B STRETCH · A

***Engineering Applications of Artificial Intelligence* (class B)**

When joined with layout extraction by segmentation: imagery → vector network → T1' → type

Class A

No natural class-A venue; not forced

SUPPLEMENTARY · CLASS C

***Journal of Ship Production and Design* (class C)**

Shipyard production organisation, the closest setting

Three more class-C venues: *Int. J. Naval Architecture and Ocean Engineering*, *Journal of Marine Science and Engineering*, *International Shipbuilding Progress*

Class B first, meeting the school's “at least one class-B-or-above journal paper”; no natural class-A venue, not forced.

References

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Data and material: thesis-core a882742 (T17–T23 results and reports) and tag manifesto-v1; library notes/occupancy_anchors_20261009.md (T24, c881609); the yard list (00_总控, 10 Oct 2026) and the review and follow-up guidance (9 Oct 2026); yard maps in 20_数据/船厂地图; OpenStreetMap contributors via Overpass (9 Oct 2026), ODbL 1.0; open imagery IGN BD ORTHO, LGLN DOP20, GSI tiles, USDA NAIP (terms checked on inclusion); Esri World Imagery (not open, local tracing only); CIMC Raffles cleaner-production notice (29 Apr 2024) and 2021 environmental disclosure; the school's 2026 journal list (23 Sep 2026).

Back to the thesis briefing, or on to the neighbouring studies

The predictor is T1' itself

One two-second linear program per yard; reservation plateaus at 36% and 35% of the system T1', each predicting the other within about 3%; three readings of the binding resource.

Key-element review is a fixed step

After T1', the author checks the top-three dual-price elements on imagery and then sets the resource granularity; in both yards the check changed the model.

Fifteen public layouts

Two done, on either side of H1; all 20 listed yards have maps, group A's six being traced; grouped by key-element type and tested leave-one-out on plateau, K^* , threshold and upgrade value.

Bound first, binding element next, simulation last: chapter 6 tests whether a two-second linear program can screen a new shipyard.

[← Back to the thesis briefing \(road-occupying transport\)](#)[Occupancy model and bounds →](#)[Capacity interval →](#)[Fleet-network co-design →](#)[The fleet-sizing \(MY-B\) briefing →](#)[The segmentation \(MY-A\) briefing →](#)

Thank you.